

```
# ALPACA MTF BOT — CLAUDE CODE SESSION HANDOFF
**Last updated:** Wednesday April 15, 2026 (~post-RTH PDT — Full file audit complete; P1 findings
**Bot location:** `~/Desktop/alpaca-mtf-bot_FINAL/`
**Prepared by:** Claude Sonnet
**Handoff to:** Claude Code — next continuation session
**File state:** All modified files syntax-verified clean. DATA-1-DATA-5 complete. P1 audit finding
**Bot status:**  RUNNING — verify PID and log tail after restart below.
```

SESSION-START PROTOCOL (MANDATORY)

Before touching any code, every session must:

1. Read `CLAUDE.md` in the project root
2. **Grep-verify** every fix listed in the prior session's handoff against actual code — not notes
3. Check `logs/mtf\_bot.log` tail for overnight errors
4. Only then begin new work

Board rule: All 26 board members at 150 IQ. Full protocol in `CLAUDE.md`.

Handoff rule: Update this file AND export PDF at the end of every session, before context runs

MANDATORY GUARDRAILS (from CLAUDE.md — full 8-guardrail version Apr 15 2026)

1. **4-tier data source hierarchy** — T1=Alpaca Data API (`data/fetcher.py` + `data/alpaca\_data.py`)
 2. **Execution isolation** — `TradingClient` in `execution/broker.py` only; `StockHistoricalDataClient` in `data/fetcher.py`
 3. **Output to `logs/` only** — lockfiles in `/tmp/` exempt; caches in `data/cache/`; runtime state in `data/state/`
 4. **RTH block:** 9:30 AM–4:00 PM ET (6:30 AM–1:00 PM PT), Monday–Friday — all analysis/backtest scripts must be RTH-aware
 5. **Data quality contract** — zero-volume reject, price sanity, timestamp freshness, source tag accuracy
 6. **Environment management** — `TRADING\_ENV` env var; `paper=True` hardcoded in broker.py (never `live=True`)
 7. **Structured logging** — all trade lifecycle events → `logs/trade\_events.jsonl` (PT timestamps, UTC timestamps)
 8. **PT everywhere** — all user-facing output in `America/Los\_Angeles`. Label: `PT` (not PST/PDT)
- Approval-gate:** Present ALL proposed changes, wait for confirmation before writing.

PROJECT OVERVIEW

- **Type:** Alpaca Markets paper trading bot — MTF (multi-timeframe) confluence scoring
- **Account:** Paper (~\$2,500 equity), PDT-constrained (max 3 day trades / 5-day rolling window)
- **Goal:** Aggressive — \$2,500 → \$25,000. Risk rules revisit when equity exceeds \$25K.
- **Scoring:** 12-point live confluence system; 16-point logged for validation only
- **Active profile:** `paper` — INTRADAY_STOP_ATR_MULT=1.25 | TARGET=2.5x | MIN_SCORE=9/12 (dynamic)
- **Kill switch:** 15%/day (BoD=3 runtime cap). Session P&L degradation: 0%→1.0x | 7.5%→0.5x | 15%→0.2x
- **PDT state:** Check current log — PDT window rolls daily.

COMPLETE FIX REGISTRY — ALL SESSIONS (code-verified)

Apr 7 Emergency Session

ID	Fix	File	Grep Verified
Fix 1	GTC cancel always-fires at RTH open (removed skip-if-unchanged)	`main.py`	✓
Fix 2	Cycle watchdog — 12-min RTH hang → Slack + os.execv()	`main.py`	✓
Fix 3	Partial close fail alerting — Slack at 2 consecutive failures	`main.py`	✓
Fix 4	Dashboard all_time P&L from EOD not trade_log	`generate_dashboard.py`	✓
Fix 5	EOD/trade_log patched for COIN manual close	`trade_log.json` + EOD files	✓ (data)

Apr 8 Session

ID	Fix	File	Grep Verified
Fix 6	Stale bar: measure from bar CLOSE not OPEN (+15 phantom min)	`main.py:992`	✓
Fix 7	Real-time entry price via fast_info.last_price (socket 4s timeout)	`main.py:965`	✓
Fix 8	yfinance timeout wrappers — `_get_weekly_bias` (8s), `_intraday_override` (4s)	`signal`	
Fix 9	Slack timestamps → PT timezone	`alerts.py:49,103`	✓
Fix 10	Weekly review "PT" label (was "PDT")	`weekly_review.py`	✓

Apr 8 Multi-Session Audit + P5 Sprint

ID	Fix	File	Grep Verified
P5-C1	FVG MultiIndex crash — `wk.columns.get_level_values(0)` + socket timeout	`signal_gener`	
P5-C2	run_movers ImportError — `TF_15M/1H/DAILY/WEEKLY` imports	`run_movers.py:27,42`	✓
P5-M1	Lockfile execv — heartbeat path now unlocks before os.execv()	`main.py`	✓
P5-M1b	RTH block added to `audit_signals.py` and `audit_final.py`	Both files	✓
P5-M2	fetch_vix Guardrail 1 — replaced raw requests.get with yfinance fast_info	`scan_to_htm`	
P5-M3	Bucket A halt exemption — `circuit_breaker=True` param, no exemption on true halt	`mai`	
P5-H1	Daily reset decoupled — buffers clear separate from network call	`main.py:5087`	✓
P5-H2	Fill price crosstalk — `submitted_after` timestamp filter	`main.py:_fetch_actual_fill`	
P5-H3	Orphan GTC 404 — confirmed clean, no code change needed	`main.py`	✓
P5-H4	Opening buffer comment corrected; config=30 confirmed	`main.py:3181`	✓
P5-L1	Loss driver badge — `"BUG IDENTIFIED" in label` (was `"MECHANICAL"`)	`weekly_review.py`	
P5-L2	Weekend expiry + empty chain guard — weekday filter + `if calls.empty`	`scan_to_html.p`	
P5-L3	FVG sync HTTP — socket 4s timeout in `_compute_fvg_mult`	`main.py:428`	✓
PDT-1	PDT centralization — `compute_rolling_pdt_count()` in portfolio_tracker	`portfolio_tra`	
PDT-2	`pdt_slots_used` excludes SYNC entries in `write_eod_summary`	`portfolio_tracker.py:19`	
PDT-3	Dashboard PDT delegate to canonical function	`generate_dashboard.py`	✓
PDT-4	Weekly review PDT delegate + holiday-aware	`weekly_review.py`	✓
DASH-1	Dashboard urllib rewrite — no alpaca-py, no MACL risk	`generate_dashboard.py:74`	✓
DASH-2	KPI reorder: Equity→All-Time P&L→Realized→Unrealized→PDT→SPY	`generate_dashboard.py`	
DASH-3	Re-entry unrealized P&L — positive filter via trade_log open	`generate_dashboard.py:3`	
WR-1	Weekly review archive rebuild — removed stale `if exists: continue`	`weekly_review.py`	
MRI-1	MRI socket timeout — `setdefaulttimeout(5.0)` wraps all `_compute()` downloads	`macro`	
RSI-1	RSI NaN divide-by-zero — `np.where(avg_loss==0, 100, ...)`	`indicators/rsi.py:21`	✓
NM-1	news_monitor `_macro_risk_window` — dict-keyed (hash-ts), prevents restart inflation	`n`	
NM-2	news_monitor `_seen_hashes` disk persistence — survives restarts	`news_monitor.py:_load`	
SC-1	score_comparison 14-day file prune	`signal_generator.py`	✓
AH-1	AH GTC retry message — PDT block vs window-closing distinction	`main.py`	✓
LOG-1	log_fill.py `--entry-premium` argument for external closes	`log_fill.py`	✓

Apr 9 Session

ID	Fix	File	Grep Verified
ALERTS-1	`send_slack(message: str)` added to `alerts.py` – was missing, ImportError silenced p		

Apr 13 Session (Nightly Audit Patches + BUG-W2 + Build #4)

ID	Fix	File	Verified
BUG-W2	`threading.Lock()` around all 5 `_last_cycle_complete` read/write sites `main.py`		
Build #4	Layer 4 MIN_SCORE gate (session loss $\geq 3\%$ $\rightarrow +1$, $\geq 7.5\%$ $\rightarrow +2$); win-side size boost (up t		
Build #5-#8	Verified already implemented: thesis invalidation exits, RVOL gate, overnight expo		
Dash P&L	Double-count guard + `unrealized_intraday_pl` for daily P&L display `generate_dashb		
Dash restore	File truncated at 585 lines – restored to 731 lines with full HTML tail + generat		
MEMO-1	Startup validation wrapped in 3-attempt retry loops, 5s backoff `main.py`	py_com	
MEMO-2	`check_partial_exits`: `_fetch_actual_fill_price(poll_secs=1.5, submitted_after=pc_ts)		
MEMO-3	RSI overwrite – clean version (redundant line removed) `indicators/rsi.py`	py_co	
MEMO-4	`live_data_writer.py` exception catch elevated to `ERROR` with `exc_info=True` `live_		
MEMO-5	`_timedelta`/`_date` aliases replaced with direct `timedelta`/`date` imports (7 call si		

Apr 14 Post-RTH Evening Session (Gemini Audit – GEM-1 through GEM-6)

ID	Fix	File	Verified
GEM-1	`_reconcile_positions` size-mismatch block now banks P&L for externally-closed shares be		
GEM-2	`inject_news_state(alert_count)` method added to `MacroRiskIndex`. Called every scan cyc		
GEM-3	`_SECTOR_MAP` expanded: ARM, TXN, KLAC, LRCX, AMAT, ON, WOLF, NET, SHOP, TTD, OKTA, GTLB		
GEM-4	`40310000` auto-recovery in `partial_close_position` and `close_position` – cancels bloc		
GEM-5	PDT streak threshold relaxed 3 $\rightarrow$ 2 consecutive 12/12 scans (all buckets). `main.py`		
GEM-6	Patch 1 PDT guard – skips emergency GTC if PDT=3/3 AND opened today (prevents 40310100 r		

Apr 14 Evening – Phantom GTC Stop Fix (GHOST-1, GHOST-2)

ID	Fix	File	Verified
GHOST-1	Patch 1 calls `get_open_position()` before submitting emergency GTC. If position absen		
GHOST-2	AH GTC loop calls `get_open_position()` before submitting GTC. Same phantom-stop fix f		

Apr 14 Evening – Weekly Review Dashboard Improvements (WRD-1–WRD-9)

ID	Change	File	Verified
WRD-1	Week P&L stat tile: `(closed only)` moved to subtitle div `weekly_review.py`		
WRD-2	"Best Performers" $\rightarrow$ "Biggest Winners"; "Worst Performers" $\rightarrow$ "Biggest Losers" `weekly_r		
WRD-3	`_week_stats()` returns full per-reason `exit_reasons` dict; exit breakdown uses live pe		
WRD-4	KPI grid 4 $\rightarrow$ 7 cols: added Max Drawdown, Avg Hold Time, ROI % (\$2.5K) `weekly_review.py`		
WRD-5	Score-Outcome table rows tinted: green $\geq 60\%$ WR, yellow 40–60%, red $< 40\%$ `weekly_review		
WRD-6	"Avg TQI" $\rightarrow$ "Avg TQI (Trade Quality Index)"; performer row `TQI {val}/100` `weekly_rev		
WRD-7	`_n<30` disclaimer fires when total_trades < 30 ; "Data as of:" PT timestamp `weekly_r		
WRD-8	Future day tiles: `opacity:0.35; pointer-events:none` when d $>$ today `weekly_review.py		
WRD-9	Overnight stat tile: lifetime overnight WR row added `weekly_review.py`		

Apr 15 Pre-RTH — DATA-1→DATA-5: Full yfinance→Alpaca Data Migration + Structured Logging

ID	Fix	Files	Verified
DATA-1	`_get_weekly_bias()`: `yf.download` → `fetch_bars(TF_WEEKLY, 14)`	`_intraday_override`	
DATA-2a-j	10 yfinance equity/ETF call sites in `main.py` → Alpaca Data T1 (bars via `fetch_bars`)		
DATA-3	`macro_risk_index._compute()`: TLT/SPY/HYG/LQD/USO/GLD/EWJ/EWG → `_alpaca_session_pct()`		
DATA-4	Created `data/fmp_client.py` — `get_earnings_dates(symbol)` via FMP T2. Replaces `yf.Ti`		
DATA-5	Created `trade_logger.py` — Guardrail-7 schema to `logs/trade_events.jsonl`. Wired into		
TZ-1a-i	9 PT display fixes across dashboard, weekly review, scanner, audit scripts	Multiple	
SCAN-VWAP	SPY 1-min VWAP → `fetch_bars(TF_1M, 400)`	`scan_to_html.py`	✓
TF-NEW	`TF_1M="1Min"`, `TF_5M="5Min"` added to config.py + TF_MAP	`config.py`, `data/fetcher`	

Apr 14 RTH — Scan Page & Options Redesign

ID	Fix	File	Verified
OPT-CSS	`options_scanner.py` — full CSS/layout redesign to match scan_results aesthetics	`op`	
DTE-CACHE	ØDTE tile `no_rec` path checks `dte_prev.json` cache; shows `LAST KNOWN` badge	`sc`	
PLN-CACHE	After computing `today_pnl = equity - last_equity`, atomically writes `logs/daily_pnl`		
PLN-READ	Bot health bar reads `daily_pnl_cache.json` first; falls back to `risk.daily_pnl` onl		

Apr 13-14 — Overnight RTH Pre-flight + Dashboard Fix

ID	Fix	File	Verified
M1	`_touch_cycle_ts()` unconditional at TOP of `while True:` loop	`main.py`	✓
M2	GTC partial fill P&L banking — `partial_pnl` and `partial_exit_price` written before `qty_r`		
M3	GTC cancel loop gate: `not is_market_open()` → `_get_tod_phase() == "closed"`	`main.py`	
B1	DAY stop race condition — pre-flight market price check before submitting RTH DAY stops	`	
B2	Pre-market mover cache empty at RTH — disk reload fallback from `logs/premarket_movers.json`		
B3	EWJ/EWG both-None — `logger.warning()` fires when both return None	`events/macro_risk_ind`	
D1	Dashboard silent API error — `logger.error()` in `_load_alpaca()` except block	`generate_`	
W-RTH	Watchdog overnight→RTH transition false-alarm — `_prev_rth` bool resets baseline at 9:30		
SH-1	`launch_bots.sh` flock-pgrep (macOS compatibility)	`launch_bots.sh`	✓
DASH-DOT	`generate_dashboard.py` dotenv — `override=True` in `_load_alpaca()`	`generate_dash`	
LW-STALE	Stale `live_data_writer.py` — killed and relaunched	`live_data_writer.py`	✓
WR-VALID	`weekly_review.py` — Live Strategy Validation panel replaces misleading 27.9% referen		
WR-LTW	Lifetime Win Rate + Lifetime P&L cells added to stats row	`weekly_review.py`	✓
ORPHAN	Cancelled orphaned SPY/QQQ GTC stop orders (no positions)	Alpaca API	✓

Apr 15 Continuation Session — External Audit P1 Fixes + Full File Audit

****Source:**** External AI audit (non-Gemini board review); all findings board-approved before implem

ID	Fix	File	Location	Board
F3	PDT phantom removal: added per-symbol names to log + hard cap at 2 removals per sync call.			
F5	AH spread gate: skip overnight entry when bid/ask spread > 2% (stale/thin AH market).	`_spr`		
F7	Per-symbol daily ATR cache with 4-hour TTL. Caches `daily_df` alongside `atr_pct` and `rvol`			
F10	Short pre-flight gate: uses `buying_power` (not `equity`) for whether shorts are currently			

****Dashboard fixes – same session (external AI board review):****

ID	Fix	File	Location	Board
DASH-KS	Kill switch gauge hardcoded at 5% → reads `config.MAX_DAILY_LOSS_PCT` dynamically.	Lab		
DASH-PT	`scan_to_html.py` timestamp used `strftime("%Z")` which returns "PDT" during DST – vio			
WR-FOOTER	`weekly_review.py` footer showed `Stop {}.ATR · Target {}.ATR` (empty braces – templ			
TB-1	Lockfile path mismatch: heartbeat restart handler used `os.unlink("/tmp/alpaca_mtf_bot.lo			

△ PENDING ITEMS – NOT YET ADDRESSED (All board-reviewed/approved)

P0 – Critical, Address Before Next RTH

ID	Item	Source	File(s)	Board Member	Notes
P0-REGIME	**SPY regime mismatch across dashboards.**	`dashboard.html`	reads `_spy_event_type`		
P0-TQI	**TQI "no history" with 15 closed trades.**	`_tqi_history`	is a module-level global in		

P1 – High Priority, Address This Session

ID	Item	Source	File(s)	Board Member	Notes
P1-PDT-NEXT	**PDT next slot date missing from `dashboard.html` KPI strip.**	`scan_results.html`			
P1-OVERNIGHT-TAG	**Overnight position tag is ambiguous.**	A 10:50 AM PT entry	tagged `OVERNIGH		
P1-PNL-AUDIT	**P&L mismatch between scan page (\$38.58) and dashboard (\$28.84 + \$10.46 = \$39.30				
P1-SCORE-TABLE	**Weekly review score table shows 100% WR on n=6 (score 10/12) – misleading wit				
P1-SCORE-CACHE	**Finding 13: `_get_live_score()` called 12–16 times redundantly per cycle.**	E			

P2 – Medium Priority, Next Sprint

ID	Item	Source	File(s)	Board Member	Notes
P2-OPTIONS-REJECT	**Options scanner shows 0 recommendations with no rejection explanation.**	0			
P2-STALE-WARN	**Dashboard has no stale-data warning banner.**	If `generate_dashboard.py` stops			
P2-MANUAL-AUDIT	**Finding 17: No audit trail for manual/external overrides.**	When positions a			
P2-MRI-SOCKET	**`macro_risk_index.py` socket timeout not restored in `finally` block in `_comp				
P2-MRI-DOCSTRING	**`macro_risk_index.py` stale-threshold docstring says ">2 hours downtime: st				
P2-BROKER-RETRY	**`submit_limit_order()` in `broker.py` has no retry logic.**	`submit_market_o			
P2-TB2	**`_cancel_and_reconcile_gtc_stops()` docstring claims "called at startup AND at the st				

P3 – Deferred / Architectural

ID	Item	Source	Status
T2	Sizing calibration – Kelly floors stress-tested against backtest data (`position-sizer.py`		
T3	Exit refinement on MRI STRESSED+: push profitable positions to breakeven CLAUDE.md backlo		
T3b	Persist macro risk window to `data/state/hybrid_state.json` (currently in-memory) CLAUDE		
DATA-4 ph2	FMP economic calendar feeds into `events/` (earnings calendar done; econ calendar d		
backtest_12pt	`yf.download(TICKERS)` bulk daily → `fetch_bars()` loop (T1 compliance) CLAUDE		
P5-L3 alt	Decoupling exits to 30s thread – requires tracker locks + global guards Prior sess		
P3-STATE-CONSOLIDATE	Three separate data sources for dashboards (`bot_status.json`, `scan_resu		

ARCHITECTURE INVARIANTS — DO NOT CHANGE WITHOUT BOARD VOTE

1. ****SPY 5-min bar-over-bar is the SOLE entry gate.**** MRI and Macro Regime Detector only adjust qu
2. ****Keywords are display-only.**** CAUTION/MONITOR = zero size impact. Only HALT = 0.0x.
3. ****PDT is a hard 3-slot rolling window.**** At PDT=3/3: GTC stop-market submitted (Alpaca blocks s
4. ****Bar staleness uses CLOSE-based age.**** `\_bar\_ts\_et + timedelta(minutes=15)` – do not revert to
5. ****Entry price = Alpaca Data real-time last trade + bar close fallback.**** `get_latest_trade(symb
6. ****Kill switch is 15% for paper.**** Revisit ONLY when equity exceeds \$25K.
7. ****Bucket A (TQQQ/SQQQ/TSLL) exemption from safe_close_all:**** Applies ONLY on routine news halts
8. ****paper=True hardcoded in broker.py.**** Change to False ONLY at live launch after full board vot
9. ****MRI is background only.**** Does not gate entries directly – sets size floor and MIN_SCORE floo
10. ****Max correlated exposure:**** No more than 2 simultaneous positions with beta correlation >0.7.
11. ****Overnight exposure budget:**** Max 100% of account equity in overnight positions.
12. ****VIX-adjusted stop widening:**** VIX > 25 → stop mult 1.5x (≤30) or 2.0x (>30). Target scales p

CURRENT ENTRY GATE STACK

Hard Blocks

Gate	Condition
Market closed	`tod_phase == "closed"`
News HALT (circuit breaker)	`news_size_mult == 0.0` → `_safe_close_all(circuit_breaker=True)`
Kill switch	Daily loss >15%
Session HALT	`_halt_entries_for_session` – resets daily
PDT=3/3	Score <10/12 or <2 confirm scans (GEM-5: was 3)

Per-Symbol Gates

Gate	Condition
Stale bar	>15 min from bar CLOSE (close-based since Fix 6)
Already has position	Alpaca confirms open
Sector correlation	Same-sector position held
R:R minimum	Stop too tight vs target
Score threshold	<9/12 (dynamic via MRI floor – up to 7 layers)
Size < \$50 notional	After all multipliers
Rule 2 gap-down	Symbol gapped ≥3% premarket → 30-min settling window
AH spread gate	AH bid/ask spread >2% → skip overnight entry (F5)

Size Reducers

Reducer	Status
SPY event (BROAD/SECTOR/EXTREME)	Active
MRI level size floor (0.40–1.00x)	Active
TOD midday size	Active
Session P&L (loss → 0.0x, win → 1.25x cap)	Active (Build #4)
Overnight exposure reducer	Active
Kill switch degradation	Linear 1.0x→0.5x→0.0x

```

### Hang Prevention
| Guard | Timeout |
|-----|-----|
| Weekly bias / intraday override (fetch_bars T1) | Alpaca SDK internal |
| MRI `_compute()` yfinance calls (^VIX/VIX3M/JPY=X) | 5s socket timeout |
| MRI `_compute()` ETF calls (fetch_bars T1) | 0.5s throttle, 5 retries |
| Entry price live fetch (`get_latest_trade` T1) | `_TIMEOUT = 4.0s` in `alpaca_data.py` |
| Cycle watchdog | 12-min RTH → Slack + os.execv() |

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## REMAINING yfinance IN BOT (all compliant)

| File | Use | Tier | Status |
|-----|-----|-----|-----|
| `strategy/volatility_regime.py` | `^VIX`, `^VIX3M` only | T4 approved |  |
| `events/macro_risk_index.py` | `^VIX`, `^VIX3M`, `JPY=X` only | T4 approved |  |
| `data/premarket.py` | `yf.screen()` screener | No T1 equivalent |  |
| `options_scanner.py`, `scan_to_html.py` | `yf.Ticker().options` chains, `.news` headlines | No T1 |
| `backtest_12pt.py` | `yf.download(TICKERS)` bulk daily history | **Flagged T1 migration** |  Si

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## RESTART PROCEDURE

```bash
cd ~/Desktop/alpaca-mtf-bot_FINAL

1. Parse-check all modified files
python3 -m py_compile main.py && echo "main OK"
python3 -m py_compile strategy/signal_generator.py && echo "signal_generator OK"
python3 -m py_compile events/macro_risk_index.py && echo "mri OK"
python3 -m py_compile execution/portfolio_tracker.py && echo "tracker OK"
python3 -m py_compile execution/broker.py && echo "broker OK"
python3 -m py_compile generate_dashboard.py && echo "dashboard OK"
python3 -m py_compile weekly_review.py && echo "weekly_review OK"
python3 -m py_compile scan_to_html.py && echo "scan_to_html OK"

2. Kill existing process
pkill -f "python.*main.py" 2>/dev/null; sleep 2

3. Start bot
nohup python3 main.py >> logs/bot_stdout.log 2>&1 &

4. Confirm startup
sleep 5 && tail -30 logs/mtf_bot.log

Expected log lines:
"Cycle watchdog armed: RTH hang threshold = 12 min"
"Position reconciliation: OK - N position(s) in sync"
"MRI refreshed: score=X level=Y"
"BoD-3: Paper kill switch capped at 15%"
```

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SESSION-START CHECKLIST (all sessions – run at top)

```bash

# Core architecture

```

grep -n "_last_cycle_lock" main.py | head -5 # BUG-W2 threading
grep -n "circuit_breaker" main.py | head -5 # P5-M3
grep -n "_pnl_min_score\|_pnl_size_mult" main.py | head -5 # Build #4
grep -n "def send_slack" alerts.py # ALERTS-1
grep -n "np.where.*avg_loss" indicators/rsi.py # RSI-1
grep -n "setdefaulttimeout" events/macro_risk_index.py | head -3 # MRI timeout
grep -n "_load_seen_hashes\|_save_seen_hashes" events/news_monitor.py # NM-2

```

# Data source migration

```

grep -n "get_latest_trade" main.py | head -8 # DATA-2b entries
grep -n "_alpaca_session_pct" events/macro_risk_index.py | head -6 # DATA-3 MRI
grep -n "get_earnings_dates" main.py # DATA-4 FMP
grep -n "log_trade_event\|_log_trade_event" main.py | head -5 # DATA-5 JSONL

```

# Gemini audit patches

```

grep -n "banking.*externally-closed\|Qty mismatch: banking" main.py | head -3 # GEM-1
grep -n "def inject_news_state" events/macro_risk_index.py # GEM-2
grep -n "40310000.*cancel\|cancel.*40310000" execution/broker.py | head -4 # GEM-4
grep -n "streak.*\//2 consecutive" main.py | head -3 # GEM-5 (was /3)

```

# Apr 15 P1 fixes (this session)

```

grep -n "_removal_cap\|_removed_syms" execution/portfolio_tracker.py # F3 PDT log
grep -n "AH spread.*> 2%\|spread_pct.*0.02" main.py # F5 spread gate
grep -n "_atr_cache\|_ATR_CACHE_TTL" main.py | head -5 # F7 ATR cache
grep -n "buying_power.*_sv_bp\|_sv_bp.*buying_power" main.py # F10 buying_power
grep -n "MAX_DAILY_LOSS_PCT.*max_loss_pct\|import config.*_cfg" generate_dashboard.py # DASH-KS k
grep -n "'PT'" scan_to_html.py | grep -v "PDT_TZ\|comment" | head -5 # DASH-PT timezone fix
grep -n "_stop_mult.*INTRADAY_STOP\|_tgt_mult.*INTRADAY_TARGET" weekly_review.py # WR-FOOTER
grep -n "_lockfile_path" main.py | grep unlink # TB-1 lockfile path
```

```

BOARD ROSTER (26 seats – all at 150 IQ)

Board of Directors (BoD) – 5 seats

Seat	Person	Domain
-----	-----	-----
Chief Investment Strategist	Jim Simons	Signal construction, regime adaptation
Chief Risk Officer	Nassim Nicholas Taleb	Tail risk, gap risk, position limits
Technology Strategy	Thomas Peterffy	Infrastructure, error handling
Market Microstructure	Albert Kyle	Order flow, adverse selection
Macro Regime Systems	David Shaw	Regime-conditional strategy switching

Advisory Board (AB) – 12 seats

Seat	Person	Domain
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```
|-----|-----|-----|
| Trade Execution | Larry Harris | Order types, slippage |
| Position Sizing | Ed Thorp | Kelly criterion |
| Compliance | Arthur Levitt | PDT rules, short sales |
| Global Macro | Paul Tudor Jones | Macro regime, overnight holds |
| Cross-Position Risk | Ray Dalio | Correlation, portfolio construction |
| Trading Psychology | Mark Douglas | Exit psychology, loss aversion |
| 0DTE / Options | Tom Sosnoff + Euan Sinclair + Dan Nathan | Gamma, IV, short-dated premium |
| Tape Reading | Peter Brandt | Tape reading, market internals |
| Momentum Factors | Jegadeesh + Titman | Momentum signal |
| Factor Investing | Cliff Asness (AQR) | Factor exposure |
| ML Signal Research | Marcos López de Prado | ML signals, overfitting |
| Alpha Research | Igor Tulchinsky (WorldQuant) | Signal validation |
```

Technical Board (TB) – 9 seats

```
| Seat | Person | Domain |
|-----|-----|-----|
| Infrastructure | Brad Katsuyama | Deployment, monitoring |
| Quant Architecture | Emanuel Derman | Code architecture |
| Security | Bruce Schneier | Secrets, key rotation |
| QA & Release | Kent Beck | TDD, regression |
| Systems Engineering | Gene Kim | DevOps, reliability |
| Algo Trading Systems | Ernest Chan | Retail/institutional algo |
| Low-Latency / Async | Yaron Minsky | Async, performance |
| Data Engineering | Wes McKinney | Bar integrity, NaN handling |
| Observability | Charity Majors | Monitoring, alerting |
```

****Total: 5 BoD + 12 AB + 9 TB = 26 seats****

LOCKED / APPROVED-BUT-NOT-YET-LIVE

```
| ID | File | Status | Condition to unlock |
|----|-----|-----|-----|
| C-2 | `execution/broker.py` | `paper=True` hardcoded | Change to `False` ONLY at live launch – f
```

WHAT TO SAY TO START THE NEXT SESSION

> "Read CLAUDE.md and CLAUDE_CODE_HANDOFF.md in ~/Desktop/alpaca-mtf-bot_FINAL/ before doing anyth